



DO-IT-YOURSELF: PROJECT

```
print(myip)

#connect to the available wifi networks
station = network.WLAN(network.STA_IF)
station.active(True)
station.connect("wifi-network-name",
               "wifi-network-password")
myip2 = station.ifconfig()
print(myip2)

def qs_parse(qs):
    parameters = {}
    ampersandSplit = qs.split("&")
    for element in ampersandSplit:
        equalSplit = element.split("=")
        parameters[equalSplit[0]] =
            equalSplit[1]
    return parameters

#create the api endpoint to control the
pin provided as parameter on API call
@app.route("/pin_ctl")
def index(req, resp):
    queryString = req.qs
```

```
parameters = qs_parse(queryString)
print(parameters)
x= int(parameters['pin'])
pin = machine.Pin(x, machine.Pin.OUT)
pin.value(0)
time.sleep(0.5)
pin.value(1)
time.sleep(0.5)
pin.value(0)
time.sleep(0.5)
pin.value(1)
time.sleep(0.5)
yield from picoweb.start_
    response(resp)
yield from resp.write("done")

if __name__ == "__main__":
    app.run(debug=True, host="0.0.0.0",
            port=80)
```

Press ctrl + x and then Y to save the code. Copy this main.py file to Node MCU board using the following syntax:

```
ampy --port /dev/ttyUSB0 put main.py
```

Connecting and operating from anywhere

Connect the LED to GPIO pin of Node MCU to test the API.

Login to the Wi-Fi homepage and check the IP address assigned to the device. Enter the address in any browser and add at the end the pin you want to control as parameter, like the example below:

```
$ http://192.168.1.14/pin_ctl?pin=3
```

To control the LED, we may connect it to GPIO 10 in the circuit diagram and run the following URL in web browser:

```
http://192.168.1.14/pin_ctl?pin=10 EFY
```



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